

★ MEMORY HIERARCHY + RAM ★

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► Meaning in one line

Registers/cache/RAM/disk , speed , cost.

► Key points

Point	Example
Hierarchy + speed/cost.	registers-fastest.
RAM + volatile + work.	running-programs.

★ IMPORTANT FOR EXAM

DRAM/ROM + PROM/cache

⚠ TRAP - read twice

Storage-RAM-mix + 'more RAM = storage' + volatility-miss

✓ **Try yourself:** Fastest + work + volatile? **Answer:** Registers; RAM.

★ DRAM/ROM + PROM/CACHE LEVELS ★

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► Meaning in one line

Dynamic/static , read-only , programmable , levels.

► Key points

Point	Example
DRAM/ROM + refresh/firmware.	-
PROM/EPROM/EEPROM + cache-L1/L2 + registers.	-

★ IMPORTANT FOR EXAM

HDD/SSD + optical/flash

⚠ TRAP - read twice

RAM-ROM-mix + 'ROM writable' + levels-miss

✓ **Try yourself:** Refresh + firmware + L1? **Answer:** DRAM; ROM; cache.

★ HDD + SSD ★

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► Meaning in one line

Spin , chip , fast , shock.

► Key points

Point	Example
HDD + spin/magnetic.	platters + cheap-big.
SSD + chip/flash + fast.	no-moving + shock-proof.

★ IMPORTANT FOR EXAM

Optical + flash + units

⚠ TRAP - read twice

SSD-HDD-mix + 'faster = bigger' + use-miss

✓ **Try yourself:** Spin + cheap-big? Chip + shock-proof? **Answer:** HDD; SSD.

★ OPTICAL + FLASH STORAGE ★

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► Meaning in one line

Light , stick.

► Key points

Point	Example
Optical + light.	CD/DVD/Blu-ray + pits.
Flash + stick.	pen/card + OTG + phones.

★ IMPORTANT FOR EXAM

Units + file-size

⚠ TRAP - read twice

Magnetic-optical-mix + 'pen = RAM' + capacity-miss

✓ **Try yourself:** Light-pits + stick? **Answer:** Optical; flash.

★ STORAGE UNITS + FILE SIZE ★

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► Meaning in one line

Bit/byte/KB/MB/GB/TB , 1024 , file.

► Key points

Point	Example
Bit/byte/KB/MB/GB/TB + 1024.	-
File-size + capacity.	movie-GB + song-MB.

★ IMPORTANT FOR EXAM

Software (system/app/utility)

⚠ TRAP - read twice

1000-1024-mix + 'MB = Mb' + file-miss

✓ **Try yourself:** 1 GB bytes? Song vs movie size? **Answer:** 1024³; MB vs GB.

★ SYSTEM + APP SOFTWARE ★

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► Meaning in one line

Platform , tools.

► Key points

Point	Example
System + platform.	Windows/Linux.
App + tools.	Word/Excel/browser.

★ IMPORTANT FOR EXAM

Utility + driver + firmware

⚠ TRAP - read twice

OS-app-mix + 'browser = OS' + use-miss

✓ **Try yourself:** Platform + tools? **Answer:** System; application.

★ UTILITY + DRIVER + FIRMWARE ★

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► Meaning in one line

Help , hardware , chip.

► Key points

Point	Example
Utility + help.	antivirus/compress/backup.
Driver/firmware + hardware/chip.	-

★ IMPORTANT FOR EXAM

Open-source + OS-functions

⚠ TRAP - read twice

Driver-OS-mix + 'firmware = software' + use-miss

✓ **Try yourself:** Help + hardware + chip? **Answer:** Utility; driver/firmware.

★ OPEN-SOURCE + OS FUNCTIONS ★

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► Meaning in one line

Free , code , manage.

► Key points

Point	Example
Open-source + free/code.	Linux/Libre.
OS-functions + manage.	process/memory/files.

★ IMPORTANT FOR EXAM

GUI + OS-awareness + files

⚠ TRAP - read twice

Free-beer + 'open = insecure' + functions-miss

✓ **Try yourself:** Free-code + manage? **Answer:** Open-source; OS-functions.

★ GUI + OS TYPES + FILES ★

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► Meaning in one line

Click , types , folders.

► Key points

Point	Example
GUI + click/icons.	windows/mouse vs CLI.
OS-types + files/folders.	path + extension.

★ IMPORTANT FOR EXAM

Process/boot + Word

⚠ TRAP - read twice

CLI-GUI-mix + 'OS = Windows' + path-miss

✓ **Try yourself:** Click + types + folders? **Answer:** GUI; OS; files.

★ PROCESS + BOOTING ★

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► Meaning in one line

Run , multitask , start.

► Key points

Point	Example
Process/multitasking.	program + threads.
Booting + POST + loader.	power→OS.

★ IMPORTANT FOR EXAM

Word (processor + extensions)

⚠ TRAP - read twice

Program-process-mix + 'boot = login' + steps-miss

✓ **Try yourself:** Run + start? **Answer:** Process; booting.

★ WORD PROCESSOR + EXTENSION ★

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► Meaning in one line

Docs , .docx.

► Key points

Point	Example
Word + docs.	MS-Word/Libre.
Extension + .docx.	-

★ IMPORTANT FOR EXAM

Formatting + setup

⚠ TRAP - read twice

Notepad-Word-mix + '.doc = .docx' + use-miss

✓ **Try yourself:** Docs + format? **Answer:** Word + .docx.

★ TEXT FORMATTING + PAGE SETUP ★

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► Meaning in one line

Fonts , margins.

► Key points

Point	Example
Formatting + fonts.	bold/italic/size/align.
Setup + paper/orientation.	A4/portrait/margins.

★ IMPORTANT FOR EXAM

Till end: find/replace + merge + compendium!

⚠ TRAP - read twice

Format-miss + 'default fine' + setup-blind

✓ **Try yourself:** Fonts + margins? **Answer:** Format + setup.